CLOXABIOTIC INJECTION
Cloxacillin Sodium BP

Product Name
CLOXABIOTIC INJECTION 250 mg
CLOXABIOTIC INJECTION 500 mg

Product Description
A white crystalline hygroscopic powder in glass vials. A clear and colorless solution after reconstituted.

Composition
CLOXABIOTIC INJECTION 250 mg
Each vial contains Cloxacillin sodium BP 250 mg

CLOXABIOTIC INJECTION 500 mg
Each vial contains Cloxacillin sodium BP 500 mg

Pharmacodynamics
Cloxacillin is a bactericidal antibiotic that is particularly useful against penicillin-producing-staphylococci. Cloxacillin is a semi-synthetic member of the penicillin family. The penicillin nucleus consists of a thiazolidine ring fused with a β-Lactam ring, to which is attached a side-chain. The side-chain determines most of the antibacterial properties of the penicillin in question. It kills bacteria by interfering in the synthesis of the bacterial cell wall. Cloxacillin binds to a penicillin-binding protein (PBPs) on the bacterial cell wall, thus interfering with peptidoglycan synthesis. Peptidoglycan is a heteropolymeric structure that provides the cell wall its mechanical stability. The final stage of peptidoglycan synthesis involves the completion of the cross-linking, where the terminal glycine residue of the pentaglycine bridge is linked to the fourth residue of the pentapeptide (d-alanine). The transpeptidase enzyme that performs this step is inhibited by penicillins and cephalosporins. As a result, the bacterial cell wall is weakened, the cell swells, and then ruptures. Cloxacillin resists the action of bacterial penicillinase probably because of the steric hindrance induced by the acyl side-chain which prevents the opening of the B-lactam ring.

Pharmacokinetics
Blood levels and tissue distribution, peak serum concentration of Cloxacillin area achieved approximately ½-1 hour after intramuscular administration. Peak serum concentrations were 6-7 and 14.7 mg/l after a 250 mg and 500 mg dose respectively.

High serum levels of the drug are achieved when Cloxacillin is administered by intravenous bolus or by short intravenous infusion modes. After an infusion of 2gm, Cloxacillin in 20 mg/l, 7.5 min, 60 min, and 4 hours, respectively, after the end of the infusion.

The degree of serum protein binding for Cloxacillin has been estimated as 94%. Therefore, through the mean free concentration of Cloxacillin, the serum level of the drug was 14 mg/l. After 2 hours the serum level has fallen 2.3 mg/l.

Toxicology
The toxicology has been investigated in mice, rabbits, rats, and dogs. These animal studies have failed to demonstrate any significant toxic effects of Cloxacillin. Except following subconjunctival injection. Administration of concentrated solutions of the drug to rabbits by

this route resulted in the development of an opacity of the cornea which persisted for at least 14 days. Cloxacillin is, therefore, contraindicated for ocular administration. In acute toxicity tests, the LD value in mice for the oral, subcutaneous, and intravenous routes were >5.0, 3.0, and 1.2 gm/kg respectively. There is no evidence that Cloxacillin is teratogenic in mice and rabbits.

Indication
Cloxacillin is indicated for the treatment of:
- Skin and soft tissue infections caused by pneumococci.
- Upper and lower respiratory tract infection.

Recommended dosage
Parenteral drug products should be inspected visually for particulate matter and discoloration, before administration, whenever solution and container permit.
Cloxacillin injection is intended for intramuscular, intravenous, intrapleural, and intra-articular use.
The recommended dosages are given below. In severe, stubborn infections, a higher dosage may be administered.

<u>Adults</u>
Intramuscular Use
250 mg every 4-6 hours.

Intravenous Use
500 mg every 4-6 hours, administered into a vein either directly or via a drip-tube for 3-4 min. More rapid administration may result in convulsive seizures.

<u>Infants and Children</u>
In infants up to 2 years of age, one-quarter of the adult dosage.
In children 2-10 years of age, half the adult dosage.

<u>Renal Insufficiency</u>
In the presence of severe renal impairment (creatinine clearance <10ml/min), a reduction in dose or extension of dose interval should be considered.

Preparation of the Solution
Before reconstitution, Cloxacillin vials should be stored in a cool dry place. Solutions for injections should be freshly prepared.
If Cloxacillin is prescribed concurrently with an aminoglycoside, the two antibiotics should not be mixed in the same syringe, intravenous fluid container, or giving set because the loss of activity of the aminoglycoside can occur under these conditions.
Cloxacillin injection should not be mixed with proteinaceous fluids such as protein hydrolysates, blood or plasma, or with intravenous lipid emulsions.

Please note: The reconstituted solution must be shaken well before use

Fill Size (mg)	Volume of Diluent	Withdrawable Volume	Nominal Concentration
For IM use: Using Sterile Water for Injection, reconstitute as follows:			
250	1.9	2.0	125
500	1.7	2.0	250
For IV Injection use: Using Sterile Water for Injection, reconstitute as follows:			
250	4.9	5.0	50
500	4.8	5.0	100

IV Infusion Entire contents of the vial to be dissolved in 5 mL Water for Injections. Further, dilute the solution in 125 to 250 mL of isotonic solutions respectively. Shake well to dissolve. Administer total contents of the vial by slow infusion over 30-40 minutes.
Immediate use of the reconstituted solution is recommended

Stability of the Reconstituted Solution

The dosing consideration; the diluents to be used, the storage temperature, and the stability recommended are:

Dosing Consideration	**Diluent**	**Stability**	
		Room Temperature 30°C ± 2°C	**Refrigerated Temperature 2°C – 8°C**
Intramuscular injection	Water for injection	30 minutes	-
Intravenous injection	Water for injection	30 minutes	-
Intravenous infusion	Initial dilution with water for injection, further diluted with 0.9% sodium chloride	24 hours	48 hours
	Initial dilution with water for injection, further diluted with Fructose 10% in water	24 hours	48 hours
	Initial dilution with water for injection, further diluted with M/6 sodium lactate in water	24 hours	48 hours
	Initial dilution with water for injection, further diluted with 5% dextrose in water	12 hours	48 hours

Incompatibility
Extemporaneous admixtures of beta-lactam antibacterials and aminoglycosides may result in substantial mutual inactivation. If these groups of antibacterials are administered concurrently, they should be administered in separate sites at least 1 hour apart. Do not mix them in the same intravenous bag, bottle, or tubing.
When aminoglycosides and penicillins are administered separately by different routes, a reduction in aminoglycosides serum concentration may occur. Usually, this is clinically significant only in patients with severely impaired renal function when the excretion of both medications is delayed.

Contraindications

Cloxacillin must not be given to patients known to be hypersensitive to Penicillin and it must not be used for the treatment of infections due to penicillinase-producing organisms.

Warnings and Precautions

Patients known to be hypersensitive to penicillin should be given an antibiotic of another class. Care is necessary if very high doses of penicillin are given especially if renal function is poor because of the risk of nephrotoxicity. Care is necessary for treating patients with syphilis because Jarisch-Herxheimer's reaction may occur. It should preferably not be given to patients with infectious mononucleosis since they are especially susceptible to Cloxacillin-induced skin rash.

Interactions with Other Medicaments

The administration of probenecid with Cloxacillin results in higher serum peak concentrations and prolongs serum concentrations of Cloxacillin. Concurrent use with allopurinol may increase the possibility of skin rash, especially in hyperuricemic patients. Antidiarrheals, antiperistaltic (for example, opiates and diphenoxylate and atropine): Penicillins may cause pseudomembranous colitis which may result in severe watery diarrhoea. Concurrent or subsequent administration of antiperistaltic antidiarrheals is not recommended since they may delay the removal of toxins from the colon, thereby prolonging and/or worsening diarrhoea. Chloramphenicol or erythromycin or sulfonamides or tetracyclines may interfere with the bactericidal effect of penicillins in the treatment of meningitis or other situations where a rapid bactericidal effect is necessary. It is best to avoid concurrent therapy. Chloramphenicol and Cloxacillin are sometimes administered concurrently in paediatric patients.

Concurrent use with contraceptive, oral, may decrease the effectiveness of oral contraceptives Stimulation of estrogen metabolism or reduction in the enterohepatic circulation of estrogen can cause menstrual irregularities, intermenstrual bleeding, and unplanned pregnancies. The patient should be advised to use an alternate method of contraception while taking any of these penicillins. Probenecid may decrease renal tubular secretion of penicillins when used concurrently.

Pregnancy and Lactation

FDA Pregnancy Category B

Reproductive studies performed in the mouse, rat, and rabbit were given Cloxacillin have revealed no evidence of impaired fertility or harm to the foetus.

Breastfeeding

Penicillins are distributed into breast milk, some in low concentrations. Although significant problems in humans have not been documented, the use of penicillins by nursing mothers may lead to sensitization, diarrhoea, candidiasis, and skin rash in the infant.

Side Effects

It is also possible that nausea, diarrhoea, digestive disorders, or skin rashes in the form of maculopapular exanthem may be seen.

The incidence of skin rash due to Cloxacillin is similar to that observed in patients suffering from infectious mononucleosis.

These adverse effects include hemolytic anaemia and neutropenia both of which might have some immunological basis, prolongation of bleeding time & defective platelet function.

Convulsion and other signs of central nervous system toxicity followed intrathecal administration.
Some patients with syphilis and other spirochaete infections may experience a Jarisch Herxheimer reaction shortly after starting treatment with penicillin, which probably due to the release of endotoxins from the killed treponemes. Symptoms include fever, chills, headache, and reaction at the site of lesions. The reaction can be dangerous in cardiovascular syphilis or where there is a serious risk of increased local damage such as optic atrophy.

Symptoms and Treatment of Overdose
The only ill-effects to be expected are nausea and vomiting, abdominal discomfort, and diarrhoea. No specific treatment is required, except for the withdrawal of the drug.

Presentation
One folded paper box contains 10 vials or 50 vials.

Storage Conditions
Store below 30°C.

Shelf Life
The injections can be used within 36 months from the date of manufacture if kept as recommended.

Registration number

CLOXABIOTIC INJECTION 250 mg -	**MAL08021484AZ**
CLOXABIOTIC INJECTION 500 mg -	**MAL08021485AZ**
	BRU18122468P

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